

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

# Chapter 21: Trigonometrical Identities

Previous year board questions, topic-wise: 2 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

## 1 The three basic identities

- Assertion (A): If  $\sin^2 A + \sin A = 1$ , then  $\cos^4 A + \cos^2 A = 1$ . Reason (R):  $1 - \sin^2 A = \cos^2 A$ . (a) A true, R false (b) A false, R true (c) both true, and R is the correct reason for A (d) both true, but R is not the correct reason for A **ICSE 2025, 1 mark**

## 2 Proving identities

- Prove that  $\frac{(\cot A + \tan A - 1)(\sin A + \cos A)}{\sin^3 A + \cos^3 A} = \sec A \operatorname{cosec} A$ . **ICSE 2025, 4 marks**

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